

Gaussian Balls Refute the Proposed Complement-Concavity Extension

Counterexample packet for arXiv:2407.20064

12 August 2026

Status: candidate full counterexample, likely valid; human review requested.

The source's final conjecture is false already for the standard Gaussian in dimension two. With $K = 6B_2^2$, $L = 2B_2^2$, and $t = 1/2$, its left side is e^{-2} , while its right side is strictly larger than $(1 - e^{-2})/4 > e^{-2}$. More generally, every absolutely continuous probability measure satisfying the conjecture's concavity premise violates its proposed conclusion for some two concentric balls.

1 The conjecture

Let $\mathcal{K}_e^n$ denote the origin-symmetric convex bodies in $\mathbb{R}^n$. The final conjecture in Langharst–Liu–Tang, *The Weighted L^p Minkowski Problem* [1], says that if a probability measure μ is $(1/n)$ -concave over $\mathcal{K}_e^n$, then for all $K, L \in \mathcal{K}_e^n$ and $t \in (0, 1)$,

$$\begin{aligned} & \mu((1-t)(\mathbb{R}^n \setminus \text{int } K) + tL) \\ & \geq \left((1-t)\mu(\mathbb{R}^n \setminus \text{int } K)^{1/n} + t\mu(L)^{1/n} \right)^n. \end{aligned} \tag{1}$$

Conjecture Let μ be a probability measure over $\mathbb{R}^n$ that is $(1/n)$ -concave over $\mathcal{K}_e^n$. Then, for every $K, L \in \mathcal{K}_e^n$ and $t \in (0, 1)$,

$$\mu((1-t)(\mathbb{R}^n \setminus \text{int}(K)) + tL) \geq \left((1-t)\mu(\mathbb{R}^n \setminus \text{int}(K))^{1/n} + t\mu(L)^{1/n} \right)^n.$$

Figure 1: The source's final conjecture (PDF page 51).

The premise means that the same inequality, without complements and with the usual Minkowski combination, is already known for pairs of symmetric convex bodies. The proposed extension replaces one convex input by its nonconvex, unbounded complement. That change creates an elementary obstruction.

2 A general ball obstruction

Write $B = B_2^n$ for the Euclidean unit ball.

Lemma 1 (Exact complement sum). *Let $a > b > 0$ and set $t = (a - b)/(a + b)$. Then*

$$(1-t)(\mathbb{R}^n \setminus \text{int}(aB)) + tbB = \mathbb{R}^n \setminus \text{int}(bB). \tag{2}$$

Proof. The choice of t gives $(1-t)a - tb = b$. If $|y| \geq (1-t)a$ and $|z| \leq tb$, the reverse triangle inequality gives $|y+z| \geq b$, proving one inclusion.

Conversely, take $|x| \geq b$. If $|x| \geq (1-t)a$, use $y = x, z = 0$. If $b \leq |x| < (1-t)a$, use $y = (1-t)a x/|x|$ and $z = x - y$. Then $|z| = (1-t)a - |x| \leq (1-t)a - b = tb$, so $x = y + z$ belongs to the left side. $\square$

Theorem 2 (Structural failure). *Let μ be an absolutely continuous probability measure on $\mathbb{R}^n$ which is $(1/n)$ -concave over $\mathcal{K}_e^n$. Then the conclusion (1) fails for some $K, L \in \mathcal{K}_e^n$ and $t \in (0, 1)$.*

Proof intuition. Choose a ball $L = bB$ carrying more than half the probability. Take a much larger concentric ball $K = aB$ and tune t so that adding tL to the scaled exterior of K produces exactly the exterior of L . The left side of (1) is then the small mass outside L , while the right side retains almost all the large mass inside L . As $a \rightarrow \infty$, the coefficient of that impossible term tends to one.

Proof. Choose $b > 0$ with $\alpha := \mu(bB) > 1/2$; absolute continuity makes $\mu(\partial(bB)) = 0$. For $a > b$, set $K = aB$, $L = bB$, and $t = (a-b)/(a+b)$. Lemma 1 makes the left side of (1) equal to $1 - \alpha$. Its right side is at least

$$(t\mu(L)^{1/n})^n = t^n \alpha. \quad (3)$$

As $a \rightarrow \infty$, $t \rightarrow 1$. Since $\alpha > 1/2$, one can choose a finite a for which $t^n \alpha > 1 - \alpha$. For that choice, (3) is strictly larger than the left side, contradicting (1). $\square$

Notice that the concavity premise is used to make μ an admissible measure for the conjecture; the failure mechanism itself needs only absolute continuity and finite total mass.

3 An explicit Gaussian counterexample

Let γ_2 be standard Gaussian measure on $\mathbb{R}^2$:

$$d\gamma_2(x) = \frac{1}{2\pi} e^{-|x|^2/2} dx.$$

Eskenazis and Moschidis proved the dimensional Gaussian Brunn–Minkowski inequality [2]: for symmetric convex bodies $A, C \subset \mathbb{R}^n$,

$$\gamma_n((1-s)A + sC)^{1/n} \geq (1-s)\gamma_n(A)^{1/n} + s\gamma_n(C)^{1/n}. \quad (4)$$

Thus γ_2 satisfies exactly the premise of (1).

Take

$$n = 2, \quad K = 6B_2^2, \quad L = 2B_2^2, \quad t = \frac{1}{2}. \quad (5)$$

Lemma 1, or a direct radius calculation $3 - 1 = 2$, gives

$$\frac{1}{2}(\mathbb{R}^2 \setminus \text{int } K) + \frac{1}{2}L = \mathbb{R}^2 \setminus \text{int}(2B_2^2). \quad (6)$$

Polar integration yields

$$\gamma_2(rB_2^2) = 1 - e^{-r^2/2}. \quad (7)$$

Consequently, the left side of (1) is e^{-2} , whereas the right side is

$$\frac{1}{4} \left(e^{-9} + \sqrt{1 - e^{-2}} \right)^2 > \frac{1}{4}(1 - e^{-2}). \quad (8)$$

Finally,

$$e > 1 + 1 + \frac{1}{2} + \frac{1}{6} = \frac{8}{3} \implies e^2 > \frac{64}{9} > 5,$$

and hence

$$e^{-2} < \frac{1}{4}(1 - e^{-2}). \tag{9}$$

Equations (8)–(9) show that the proposed right side is strictly larger than the left side. All bodies are origin-symmetric convex bodies, and (4) verifies the only non-elementary hypothesis.

4 Verification, scope, and novelty

The counterexample is a full negative answer to the displayed source conjecture, not to the weighted L^p Minkowski existence theorems proved elsewhere in the paper. It also does not challenge Gaussian $(1/n)$ -concavity on symmetric convex bodies: one input in (1) is the complement of such a body and is neither convex nor bounded.

Review should focus on the exact set identity (2), the Gaussian tail formula (7), and the direction of the strict comparison (9). The accompanying script checks the radius, exact elementary certificate, and numerical gap.

On 12 August 2026, the four run indexes and bounded exact/current searches for the arXiv id, title, exact complement formula, and later citations found no existing proof or refutation. The current source version still states the conjecture on page 51. Novelty confidence is moderate-high.

Human-review recommendation. Review as a full counterexample. The example is explicit, two-dimensional, uses the canonical measure motivating the paper, and extends to every absolutely continuous probability measure satisfying the stated premise.

References

- [1] D. Langharst, J. Liu, and S. Tang, *The Weighted L^p Minkowski Problem*, arXiv:2407.20064, Appendix conjecture on PDF page 51.
- [2] A. Eskenazis and G. Moschidis, *The dimensional Brunn–Minkowski inequality in Gauss space*, J. Funct. Anal. 280 (2021), Paper No. 108914; arXiv:2004.07146.